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SAINTGITS COLLEGE OF ENGINEERING (AUTONOMOUS)
M.Tech. SYLLABUS 2026

COURSE DESCRIPTION

Course Code	26SE103	Course Credit: 4
Course Name	Database Systems and Distributed Data Management	
Course Delivery Mode	TiC & PiL	
L-T-P-H	3-0-2-0	
Pre - requisite	Nil	
Year of Introduction and Syllabus Version	2026 V1.0	

COURSE OBJECTIVES

	Cognitive Level	Knowledge Category
1 To gain knowledge of core concepts, architectures, and challenges in distributed computing environments.	L2	K1
2 To learn the principles of big data and understand how Hadoop processes and manages large - scale data.	L2	K2
3 To study HDFS and MapReduce for efficient distributed storage and parallel data processing	L3	K3
4 To learn different types of NoSQL databases and their role in handling scalable, distributed data.	L3	K3
5 To implement CRUD operations, optimize queries, and integrate NoSQL databases with applications.	L4	K4

COURSE OUTCOMES (COs)

After the successful completion of the course students will be able to:

- CO1 Understand and explain the principles, architectures, and challenges of distributed systems.
- CO2 Understand big data concepts and analyze the Hadoop ecosystem and its components.
- CO3 Apply HDFS and MapReduce techniques for distributed storage and data processing.
- CO4 Analyze NoSQL database models and compare them with traditional relational databases.
- CO5 Design and implement NoSQL database solutions with efficient querying and scalability.

CO - PO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	-	2	-	-	-
CO2	2	-	3	-	-	-
CO3	3	2	3	-	2	2
CO4	2	2	3	-	-	-
CO5	3	2	3	-	2	2

Correlation levels: 1 - Low; 2 - Medium; 3 - High; No Correlation - " - "

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COURSE IMPLEMENTATION CLOCK (IN HOURS)	1 Hr
Zeroth Module	60 Hrs
Course content delivery hours	
Internal Test	3 Hrs
(2 Assessment Tests of one and a half hour duration)	

COURSE CONTENTS

Zeroth Module: General Introduction to the Course

- Course introduction with a thorough briefing of Course Objectives and Course Outcomes - An overview of the POs that are mapped to the course	No ESE	1 Hr.
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Module - I

Module Title Big Data and Hadoop Ecosystem

Introduction to distributed database - Communication in distributed systems (RPC, RMI, message passing) - Big data concepts and characteristics - Introduction to Hadoop ecosystem - Hadoop architecture and components - Hadoop cluster setup and configuration - Hadoop YARN architecture - Hadoop ecosystem tools - Limitations of traditional systems vs Hadoop.

COs Hrs.

1,2 12

Module - II

Module Title HDFS and MapReduce Programming

Data storage and processing in Hadoop - Hadoop distributed file system architecture - HDFS components - Data replication and fault tolerance - Introduction to MapReduce paradigm - MapReduce workflow and components - Writing MapReduce programs (Java/Python) - Input and output formats - Combiner.

COs Hrs.

3 10

Module - III

Module Title NoSQL Database Programming and Applications

Introduction to NoSQL databases - Need for NoSQL - Types of NoSQL databases - Key value stores - Document stores - Column - family stores - Graph databases - Data modelling in NoSQL - Popular NoSQL databases (MongoDB, Cassandra) - CRUD operations in NoSQL databases - Querying and indexing techniques - Schema design and optimization - Working with MongoDB (documents, collections, aggregation framework) - Programming with Cassandra (CQL basics)

COs Hrs.

4, 5 14

List of Exercises

(Minimum 8 exercises to be completed)

		CO
1	Sending message from client to server through RPC in Java	1
2	Hadoop single node and multi node installation	2
3	Familiarization of HDFS commands	3

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4	To demonstrate the usage of combiners and practitioner in Hadoop	3
5	Understand NoSQL concepts and install a database.	4
6	Perform Create, Read, Update, Delete operations. (CRUD in MogoDB)	5
7	Create and query tables in Apache Cassandra.	4
8	Perform MapReduce in MongoDB.	5
9	Develop a NoSql Data Model for a specific Application	5
10	Implement the NoSQL Application in a suitable model.	5

LABORATORY MANUAL

An internally prepared laboratory manual contains

- Vision and Mission of the institution
- Details of laboratory equipment
- Details of laboratory exercises

TEXT BOOKS

- 1 G. Coulouris, J. Dollimore, T. Kindberg, and G. Blair, *Distributed Systems: Concepts and Design*, 5th Edition, Wesley, 2011, ISBN: 978 - 0132143011.
- 2 T. White, *Hadoop: The Definitive Guide*, 4th edition, O'Reilly Media, 2015, ISBN: 978 - 1491901632.
- 3 M. Fowler and P. Sadalage, *NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence*, Wesley, 2012, ISBN: 978 - 0321826626.

REFERENCE BOOKS / WEB REFERENCES

- 1 M. Kleppmann, *Designing Data - Intensive Applications*, 1st edition, O'Reilly Media, 2017, ISBN: 978 - 1449373320.
- 2 J. Leskovec, A. Rajaraman, and J. D. Ullman, *Mining of Massive Datasets*, 2nd ed, Cambridge University Press, 2014, ISBN: 978 - 1107077232.
- 3 L. George, *HBase: The Definitive Guide*, Sebastopol, O'Reilly Media, 2011, ISBN: 978 - 1449396107.
- 4 E. Hewitt and J. Carpenter, *Cassandra: The Definitive Guide*, 2nd Edition, O'Reilly Media, 2016, ISBN: 978 - 1491933664.
- 5 K. Chodorow, *MongoDB: The Definitive Guide*, 2nd Edition, O'Reilly Media, 2013, ISBN: 978 - 1449344689.

MODES OF EVALUATION**SCORE WEIGHTAGE / SPLIT MARKS****Continuous Internal Evaluation (CIE)**

50

Internal Test - I 7.5

Internal Test - II 7.5

Taskworks (Assignments/MCQ/Quiz) 10

Embedded Lab 25

End Semester Examination (ESE)

50

Total

100

End Semester Examination Pattern

The question paper consists of two parts:

Part A: 14 marks

Part B: 36 marks

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Part A contains a total of 7 questions, 2 questions each from first 2 modules and 3 questions from 3rd module, having two marks for each question. Part B contains 2 questions from each module of which student should answer any one. Each question can have a maximum 2 subdivisions and carry 12 marks

APPROVAL SUMMARY

Number and Date

Recommended by BoS

SCE/BoS/CS/M.Tech./SE/013/25- 26, 22-05-26

Approved by Academic Council

AC/RES/019/25-26, 15-06-2026

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Department of Computer Science and Engg. / M.Tech . Computer Science and Systems Engg.

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